

**FIFTH SEMESTER U.G. DEGREE (SPECIAL) EXAMINATION
NOVEMBER 2020**

(CUCBCSS—UG)

Mathematics

MAT 5B 08—DIFFERENTIAL EQUATIONS

Time : Three Hours

Maximum : 120 Marks

Section A

*Answer all questions.
Each question carries 1 mark.*

1. Write down the differential equation whose solution is $y = c_1 e^t + c_2 e^{-t}$.
2. Find that function given by the Wronkian $W[t, t^2]$?
3. Find the integrating factor of $(x^2 - 1) \frac{dy}{dx} + 3y = x$?
4. Find the complimentary function corresponding to $y'' + 2y' + y = t$.
5. Compute $\mathcal{L}\{e^{\mu t} \cos^2 t\}$.
6. What is one dimensional heat equation ?
7. Find $\mathcal{L}^{-1}(1)$.
8. Find the fundamental solutions of $y'' + 25y = t$.
9. Find the value of b_n in the Fourier series expansion of 2π -periodic function $f(x) = x^2, x \in [-\pi, \pi]$.
10. Show that derivative of an even function is odd.

Turn over

11. What do you mean by an exact differential equation ? Give an example.

12. The order of the differential equation $\frac{d^4 y}{dx^4} - \left(\frac{dy}{dx}\right)^7 = 10 + \frac{d^5 y}{dx^5}$ is

(12 × 1 = 12 marks)

Section B

*Answer at least **eight** questions.*

Each question carries 6 marks.

All questions can be attended.

Overall Ceiling 48.

13. Convert $y'' + y = 0$ into a system of first order equations.
14. Find the Fourier sine series for the 2π -periodic, function $f(x) = -x$, $x \in [-\pi, \pi]$.
15. Find the integrating factor for $ydx - xdy = 0$.
16. Find the inverse Laplace transform of $\log((s-a)/(s-b))$.
17. Define unit step function and find its Laplace transform.
18. Solve : $t^2 y'' - 2ty' - 3y = 0$.
19. Show that the inverse Laplace transform is linear.
20. Write the existence and uniqueness theorem for first order differential equations with the assumptions involved therein.
21. State Abel's theorem.
22. Solve : $\frac{dy}{dx} = (2x - 3y + 1)^2$.
23. Evaluate $\mathcal{L}(te^t \sin 2t)$.

24. Find the second order p.d.e. for which $u = f(x + at) + g(x - at)$ is a solution.
25. Find $\mathcal{L}[\cosh^2 2t]$.
26. Solve $y' + 2y = 0$ using Laplace transform.

(8 × 6 = 48 marks)

Section C

Answer at least five questions.

Each question carries 9 marks.

All questions can be attended.

Overall Ceiling 45.

27. Express the function $f(t) = \begin{cases} \sin t, & \text{if } 0 \leq t < \pi/2, \\ \cos t, & \text{if } \pi/2 \leq t < \pi \\ 0 & \text{elsewhere.} \end{cases}$ in terms of combination of unit step functions

and hence find its Laplace transform.

28. Use method of convolution to find the Laplace inverse of $\frac{1}{(s-1)^3}$.
29. Evaluate the Laplace inverse transforms of $\cot^{-1}(s/a)$ and $\frac{1}{(s^2 - 5s + 6)^2}$.
30. Derive the Euler formula for finding a_0 in the Fourier expansion of a 2π -periodic function $f(x)$.
31. State the conditions for the existence of Laplace transform of a function $f(t)$ and prove the same.
32. Find the Fourier cosine series for the function $f(t) = |t - \pi|, t \in [0, \pi]$.
33. Find the solution of the heat conduction problem : $36u_{xx} = u_t, 0 < x < 1, t > 0 ; u(0, t) = 0, u(1, t) = 0, t > 0 ; u(x, 0) = \cos(2\pi x) - \cos(5\pi x), 0 \leq x \leq 1.$
34. Get a formula for $\mathcal{L}(f(t))$ where $f(t)$ is a periodic function of period T.
35. State and prove convolution theorem for Laplace transforms.

(5 × 9 = 45 marks)

Turn over

Section D

*Answer any one question.
Each question carries 15 marks.*

36. (a) Apply method of variation of parameters to solve : $y'' - y = \sec t$.
- (b) Solve $(2x + y + 1)dx + (x - 3y + 2)dy = 0$.
37. (a) Solve the following differential equation in two ways, one of them must be using Laplace transform. If $y'' - 2y = t$, $y(0) = 1$, $y(1) = 0$.
- (b) Find the Half range Fourier sine series of $f(x) = x^2$, $x \in [0, 3]$.
38. Use method of separation of variables and solve the one dimensional heat equation completely. State the assumptions involved therein explicitly.

(1 × 15 = 15 marks)

**FIFTH SEMESTER U.G. (CUCBCSS—UG) DEGREE [SPECIAL]
EXAMINATION, NOVEMBER 2020**

Mathematics

MAT 5B 07—BASIC MATHEMATICAL ANALYSIS

(Multiple Choice Questions for SDE Candidates)

Time : 15 Minutes

Total No. of Questions : 20

Maximum : 20 Marks

INSTRUCTIONS TO THE CANDIDATE

1. This Question Paper carries Multiple Choice Questions from 1 to 20.
2. The candidate should check that the question paper supplied to him/her contains all the 20 questions in serial order.
3. Each question is provided with choices (A), (B), (C) and (D) having one correct answer. Choose the correct answer and enter it in the main answer-book.
4. The MCQ question paper will be supplied after the completion of the descriptive examination.

MAT 5B 07—BASIC MATHEMATICAL ANALYSIS

(Multiple Choice Questions for SDE Candidates)

1. If $A_n = \{n, n+1, n+2, \dots\}$, then, $\bigcap_{n=1}^{\infty} A_n = \dots$
- (A) 1. (B) $\emptyset$.
(C) ∞ . (D) n .
2. If $A = \{1, 2, 3\}$ and $B = \{4, 5\}$ which of the following is not a member of $A \times B$.
- (A) (1,4). (B) (2, 5).
(C) (3, 4). (D) (4, 3).
3. Which of the following subset of $A \times A$ defines a function on $A = \{x \in \mathbb{R} : -1 \leq x \leq 1\}$.
- (A) $C = \{(x, y) : x^2 + y^2 = 1\}$. (B) $C = \{(x, y) : x + y^2 = 1\}$.
(C) $C = \{(x, y) : x^2 + y = 1\}$. (D) None of these.
4. Which of the following set is *not* countable ?
- (A) $\{1, 2, \dots, n\}$. (B) The set $\mathbb{N}$ of natural numbers.
(C) The set $\mathbb{Q}$ of rational numbers. (D) The interval $(0, 1)$.
5. The number of injections from $S = \{1, 2\}$ to $T = \{a, b, c\}$ is ———.
- (A) 2. (B) 4.
(C) 6. (D) 8.
6. If $a \in \mathbb{R}$ such that, $0 \leq a < \varepsilon$ for every $\varepsilon > 0$ then, :
- (A) $a > 0$. (B) $a \neq 0$.
(C) $a = 0$. (D) None of these.
7. The binary representation of $3/8$ is, :
- (A) 0.0111111..... (B) 0.0101000.....
(C) 0.1011111..... (D) 0.0101111.....
8. If $0 < b < 1$, $\lim (b^n)$ equal to :
- (A) 0. (B) 1.
(C) b . (D) ∞ .
9. Limit of the sequence $\left(\frac{3n+2}{2n+1}\right)$ is ———.
- (A) 3. (B) $1/2$.
(C) 2. (D) $3/2$.

10. The smallest value of $K(\varepsilon)$ corresponding to $\varepsilon = .01$ for the sequence $\left(\frac{1}{n}\right)$ is _____.

- (A) 10. (B) 50.
(C) 100. (D) 101.

11. Which of the following is false ?

- (A) If (x_n) is a convergent sequence then $\left(x_n^2\right)$ is convergent.
(B) If (x_n) is a convergent sequence, and $x_n \geq 0$ for every n , then $(\sqrt{x_n})$ is convergent.
(C) If $\left(x_n^2\right)$ is a convergent sequence then (x_n) is convergent.
(D) If (x_n) is a convergent sequence then $\left(x_n^3\right)$ is convergent.

12. If $0 < a < b$, then $\lim \frac{a^{n+1} + b^{n+1}}{a^n + b^n}$ is :

- (A) b . (B) a .
(C) $a + b$. (D) ∞ .

13. The limit of the sequence defined inductively by, $x_1 = 1$ and $x_{n+1} = 2 + \frac{1}{x_n}$ is :

- (A) $1 - \sqrt{2}$. (B) $1 + \sqrt{2}$.
(C) $2 + \sqrt{2}$. (D) $2 - \sqrt{2}$.

14. Which of the following sequences with n^{th} term x_n diverges ?

- (A) $x_n = 1 - \frac{(-1)^n}{n}$. (B) $x_n = \frac{1 - (-1)^n}{n}$.
(C) $x_n = 1 - (-1)^n + \frac{1}{n}$. (D) $x_n = \frac{(-1)^n (n+1)}{n^2 + 1}$.

31. A ball with mass 0.15 kg. is thrown upward with initial velocity 20 m/s from the roof of a building 30 m. high. Neglect air resistance (a) Find the maximum height above the ground that the ball reaches ; (b) Assuming that the ball misses the building on the way down, find the time that it hits the ground.
32. Find the Fourier cosine series for the function $f(t) = \pi|t - \pi|, t \in [0, \pi]$.
33. Find the solution of the heat conduction problem :
- $$25u_{xx} = u_t, 0 < x < 1, t > 0; u(0, t) = 0, u(1, t) = 0, t > 0; u(x, 0) = \sin(2\pi x) - \sin(5\pi x), 0 \leq x \leq 1.$$
34. State and prove Convolution theorem for Laplace transforms.
35. Evaluate (i) $L^{-1}\left(\frac{1 - e^{-s}}{s}\right)$ and (ii) get a formula for $L(f(t))$ where $f(t)$ is a periodic function of period T .

(6 × 7 = 42 marks)

Part D

*Answer any two questions.
Each question carries 13 marks.*

36. (a) Use method of separation of variables and solve the one-dimensional heat equation completely. State the assumptions involved therein explicitly.
- (b) Find the solution of the p.d.e. $\frac{\partial^2 u}{\partial y \partial x} = 2x$.
37. (a) Solve the following differential equation in two ways, one of them must be using Laplace transform. $4y'' - y = t, y(0) = 1, y(1) = 0$.
- (b) Find the Fourier series of $f(x) = x^2, x \in [-2, 2]$ treating it as a periodic function of period 4.
38. (a) Apply method of variation of parameters to solve : $y'' - y = \sec t$.
- (b) Solve $(2x + y + 3) dx + (x - 3y + 2) dy = 0$.

(2 × 13 = 26 marks)

FIFTH SEMESTER B.A./B.Sc. DEGREE EXAMINATION, NOVEMBER 2021**(CUCBCSS—UG)****Mathematics****MAT 5B 07—BASIC MATHEMATICAL ANALYSIS****(Multiple Choice Questions for SDE Candidates)****Time : 15 Minutes****Total No. of Questions : 20****Maximum : 30 Marks****INSTRUCTIONS TO THE CANDIDATE**

1. This Question Paper carries Multiple Choice Questions from 1 to 20.
2. The candidate should check that the question paper supplied to him/her contains all the 20 questions in serial order.
3. Each question is provided with choices (A), (B), (C) and (D) having one correct answer. Choose the correct answer and enter it in the main answer-book.
4. The MCQ question paper will be supplied after the completion of the descriptive examination.

MAT 5B 07—BASIC MATHEMATICAL ANALYSIS

(Multiple Choice Questions for SDE Candidates)

1. If $A_n = \{n, n+1, n+2, \dots\}$, then $\bigcap_{n=1}^{\infty} A_n =$ _____.
- (A) 1. (B) ϕ .
- (C) ∞ . (D) n .
2. Consider the function $f(x) = \frac{1}{x^2}, x \neq 0$. Determine the image $f(E)$ where $E = \{x \in \mathbb{R} : 1 \leq x \leq 2\}$.
- (A) $[1/4, 1]$. (B) $[1/2, 1]$.
- (C) $(0, 1/4]$. (D) $[0, 1/4]$.
3. If $f(x) = 2x$, and $g(x) = 3x^2 - 1$, then $(f \circ g)(x)$ is _____.
- (A) $12x^2 - 1$. (B) $12x^2 - 2$.
- (C) $6x^2 - 1$. (D) $6x^2 - 2$.
4. Which of the following is true ?
- (A) $(f \circ g)^{-1}(H) = g^{-1}(f^{-1}(H))$. (B) $(f \circ g)^{-1}(H) = f^{-1}(g^{-1}(H))$.
- (C) $(f \circ g)^{-1}(H) \subseteq g^{-1}(f^{-1}(H))$. (D) $(f \circ g)^{-1}(H) \subseteq f^{-1}(g^{-1}(H))$.
5. For each $n \in \mathbb{N}$ let $A_n = \{(n+1)k : k \in \mathbb{N}\}$. Then, $A_1 \cap A_2$ is :
- (A) A_3 . (B) A_4 .
- (C) A_5 . (D) A_6 .
6. The function $f : \mathbb{R} \rightarrow (-1, 1)$ defined by $f(x) = x/\sqrt{x^2 + 1}$ is :
- (A) A surjection but not injection. (B) An injection but not surjection.
- (C) Neither injection nor surjection. (D) A bijection.

7. Which of the following set is not countable ?

- (A) $\{1, 2, \dots, n\}$ (B) The set $\mathbb{N}$ of natural numbers.
 (C) The set $\mathbb{Q}$ of rational numbers. (D) The interval $(0,1)$.

8. If $S = \{2, 3, 4\}$ the number of elements in $P(S)$, the power set of S , is :

- (A) 3. (B) 6.
 (C) 8. (D) 9.

9. Which of the following is *not* true ?

- (A) If $a > b$ and $c > 0$, then $a + c > b + c$.
 (B) If $a > b$ and $c < 0$, then $a + c < b + c$.
 (C) If $a > b$ and $c > 0$, then $ac > bc$.
 (D) If $a > b$ and $c < 0$, then $ac < bc$.

10. If $\alpha \in \mathbb{R}$ such that, $0 \leq \alpha < \varepsilon$ for every $\varepsilon > 0$ then, :

- (A) $\alpha > 0$. (B) $\alpha \neq 0$.
 (C) $\alpha = 0$. (D) None of these.

11. Let $S = \{x \in \mathbb{R} : x < 2\}$. Then :

- (A) Neither $\sup S$ nor $\inf S$ exist. (B) Both $\sup S$ and $\inf S$ exist.
 (C) $\sup S$ exists but less than 2. (D) $\sup S$ equal to 2.

12. If $S = \left\{1 - \frac{(-1)^n}{n}, n \in \mathbb{N}\right\}$, then :

- (A) $\sup S = 2$, $\inf S = 1/2$. (B) $\sup S = 2$, $\inf S = 0$.
 (C) $\sup S = 1$, $\inf S = 1/2$. (D) $\sup S = 1$, $\inf S = 0$.

13. The binary representation of $3/8$ is :

- (A) 0.0111111... (B) 0.0101000...
 (C) 0.1011111... (D) 0.0101111...

14. The rational number represented by $7.3141414\ldots$ is :

- (A) $7245/990$. (B) $7249/990$.
(C) $7241/990$. (D) $7243/990$.

15. The sixth term of the Fibonacci sequence is _____.

- (A) 5. (B) 6.
(C) 8. (D) 13.

16. Limit of the sequence $\left(\frac{3n+2}{2n+1}\right)$ is _____.

- (A) 3. (B) $1/2$.
(C) 2. (D) $3/2$.

17. The smallest value of $K(\varepsilon)$ corresponding to $\varepsilon = .01$ for the sequence $\left(\frac{1}{n}\right)$ is _____.

- (A) 10. (B) 50.
(C) 100. (D) 101.

18. Which of the following is false ?

- (A) If (x_n) is a convergent sequence then (x_n^2) is convergent.
(B) If (x_n) is a convergent sequence, and $x_n \geq 0$ for every n , then $(\sqrt{x_n})$ is convergent.
(C) If (x_n^2) is a convergent sequence then (x_n) is convergent.
(D) If (x_n) is a convergent sequence then (x_n^3) is convergent.

19. The limit of the sequence $\left(1 + \frac{1}{2n}\right)^n$ is :

- (A) 1. (B) ∞ .
(C) e . (D) $\sqrt{e}$.

20. The sequence $\left(4, -2, 0, 1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots \dots \dots\right)$

- (A) Monotone decreasing. (B) Monotone increasing.
(C) Ultimately Monotone decreasing. (D) Ultimately Monotone increasing.

FIFTH SEMESTER B.A./B.Sc. DEGREE EXAMINATION, NOVEMBER 2021

(CUCBCSS-UG)

Mathematics

MAT 5B 07—BASIC MATHEMATICAL ANALYSIS

Time : Three Hours

Maximum : 120 Marks

Part A*Answer all questions.**Each question carries 1 mark.*

1. Fill in the blanks : Supremum of the set $S = \{1 - 1/n; n \in \mathbb{N}\}$ is _____.
2. Determine the set $A = \{x \in \mathbb{R} : |x - 1| < |x|\}$.
3. The set of all real numbers which satisfy the inequality $0 \leq b < \epsilon, \forall \epsilon > 0$, then $b =$ _____.
4. Fill in the blanks : The Supremum property of $\mathbb{R}$ states that _____.
5. State the Trichotomy Property of $\mathbb{R}$.
6. Give the condition for a subset of $\mathbb{R}$ to be an interval of $\mathbb{R}$.
7. State the general Arithmetic Geometric mean inequality of real numbers.
8. Fill in the blanks : The characterization theorem of open sets states that _____.
9. State the Bernoulli's inequality.
10. If $a > 0$, then $\lim(a^{1/n}) =$ _____.
11. Fill in the blanks : $\text{Arg}(-2\pi) =$ _____.
12. Fill in the blanks : The Exponential form of $-1 - i =$ _____.

(12 × 1 = 12 marks)

Part B*Answer any ten questions.**Each question carries 4 marks.*

13. Define Supremum and Infimum of a set. Find them for the set $S = \{1/2^m - 1/3^n; m, n \in \mathbb{N}\}$.
14. Show that there doesn't exist a rational number r such that $r^2 = 3$.
15. If $a, b \in \mathbb{R}$, then prove that $||a| - |b|| \leq |a - b|$.
16. Prove that a real sequence can have at most one limit.

Turn over

17. If $x \in \mathbb{R}$ then prove that there exists $n_x \in \mathbb{N}$ such that $x < n_x$.
18. Discuss the convergence of the following sequences $X = (x_n)$, defined by (a)
- $$x_n = \left(1 + \frac{1}{n+1}\right)^{n-1}$$
19. Show directly that a bounded monotonic sequence is a Cauchy sequence.
20. Define Cauchy sequence. Test whether $(1/n)$ is a Cauchy sequence or not.
21. Show by an example that intersection of infinitely many open sets in $\mathbb{R}$ need not be open.
22. Discuss the convergence of $X = (x_n)$ define by $x_n = n$, if n odd and $x_n = 1/n$, if n even.
23. Show that every bounded sequence of real numbers has a converging subsequence.
24. Test the convergence of the sequence $\left(\frac{\cos n}{n}\right)$.
25. Express the complex number $(\sqrt{3} + i)^7$ in exponential form.
26. Find the principal value of $(-8i)^{\frac{1}{3}}$.

(10 × 4 = 40 marks)

Part C

*Answer any six questions.
Each question carries 7 marks.*

27. Prove that the set $\mathbb{R}$ of real numbers is uncountable.
28. State and prove the Ratio Test for the convergence of real sequence.
29. Discuss the convergence of the following sequences $X = (x_n)$, defined by
- $$(a) x_n = \left(1 + \frac{1}{n+1}\right)^{n-1} \text{ and } (b) x_n = \left(\frac{1-2}{n}\right)^n.$$
30. $X = x_n$ and $Y = y_n$ be sequences of real numbers converges to x and y respectively, then prove that $X \cdot Y$ converges to xy .
31. (a) Give an example of a convergent sequence (x_n) of positive real numbers with
- $$\lim \left(\frac{x_{n+1}}{x_n}\right) = 1.$$
- (b) Give an example of a divergent sequence (x_n) of positive real numbers with
- $$\lim \left(\frac{x_{n+1}}{x_n}\right) = 1.$$

(c) Give your comments about the property of the sequence (x_n) of positive real numbers

$$\text{with } \lim \left(\frac{x_{n+1}}{x_n} \right) = 1.$$

32. If $X = (x_n)$ is a real sequence and $X_m = (x_{m+n} : n \in \mathbb{N})$ is the m -tail of X ; $m \in \mathbb{N}$, then show that X_m converges to x if and only if X converges to x .
33. Let $X = (x_n)$ be a bounded sequence of real numbers and $x \in \mathbb{R}$ has the property that “every converging subsequence of $X = (x_n)$ converges to x ”. Prove that $X = (x_n)$ converges to x .
34. Prove or disprove the following statement : $\|z_1\| - \|z_2\| \leq |(z_1)| - |(z_2)|, \forall z_1, z_2 \in \mathbb{C}$.
35. Test the convergence of (x_n) defined by $x_n = 1 + 1/2 + 1/3 + \dots + 1/n$.

(6 × 7 = 42 marks)

Part D

*Answer any two questions.
Each question carries 13 marks.*

36. Show that there exists a positive real number x such that $x^2 = 2$.
37. (a) If $I_n = [a_n, b_n], n \in \mathbb{N}$ is a nested sequence of closed and bounded intervals, then prove that there exist a common point in every I_n .
- (b) Test the convergence of (x_n) defined by $x_n = 1 + 1/2! + 1/3! + \dots + 1/n!$.
38. (a) Define a closed set and “cluster point” of a set. Give examples for each of them.
- (b) Prove that a subset of $\mathbb{R}$ is closed in $\mathbb{R}$ if and only if it contains all of its cluster points.

(2 × 13 = 26 marks)